ROBERT PEREZ

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Portfolio: ripdesigns.com

EDUCATION

University of Colorado Boulder - College of Engineering and Applied Science **Expected May 2028**
B.S. - Creative Technology and Design
Cumulative GPA: 3.5

- Coding, Graphic Design, 3D Design, Product Design

EXPERIENCE

ATLS Institute

Creative Technology & Design Student — ***Aug 2024 – Present***

- Designed 15+ posters, infographics, and product mockups using Adobe Creative Suite, delivering visual assets for class exhibitions and peer-reviewed showcases.
- Conducted user-centered research studies on student behavior across CU Boulder and partner institutions, collecting and analyzing survey data to inform design decisions
- Collaborated with multidisciplinary student teams to brainstorm concepts, prototype solutions, and present design proposals during iterative critique sessions

BTU Lab

Lab Member — ***Aug 2024 – Present***

- Prototyped interactive hardware projects using Arduino, and ESP32 platforms
- Fabricated physical enclosures and components using 3D printers and laser cutters
- Collaborated with lab members to develop, test, and troubleshoot electronic circuits and creative technology prototypes

PROJECTS

Alarm Clock You Can't Ignore - Arduino + 3D Printing

- Built an interactive alarm clock that requires sustained physical motion to disable, encouraging user engagement through tactile interaction
- Designed and assembled a custom Arduino circuit integrating a tilt sensor to detect shaking input
- Programmed timing and sensor-based logic to control alarm activation and shutdown behavior
- Integrated MP3 player and speaker activated by specific programmed timing and deactivated with tilt sensor
- Prototyped and fabricated custom enclosures using 3D printing and laser cutting to house electronics and wiring

AI-Powered Music Recommendation Machine - ESP32 + API Integration

- Developed a physical music recommendation device using an ESP32 microcontroller, LCD display, and rotary encoder interface
- Integrated API-based AI song recommendation to generate customized music suggestions based on user-selected inputs and designed and wired custom circuitry and printed real-time recommendations using a thermal printer
- Combined creative coding, hardware prototyping, and user experience design to create an interactive entertainment system

NOTABLE SKILLS

- **Design & Software:** Photoshop, Illustrator, InDesign, Adobe Creative Suite
- **Prototyping & Hardware:** Arduino, ESP32, 3D Printing, Laser Cutting, Circuit Prototyping
- **Programming:** Python, Creative Coding, HTML, CSS, JavaScript
- **Design Practices:** Typography, Layout Design, Branding, Product Design, User Experience Design